Aveen Dayal

AI Research Scientist | Multimodal and Generative AI | Efficient Inference and Model Evaluation

aveendayal97@gmail.com | +91 9650150648 | LinkedIn | Research Website | GitHub | Google Scholar

Professional Summary

AI researcher with a Ph.D. in Artificial Intelligence from IIT Hyderabad, where I was a Prime Minister's Research Fellow and received the IIT Hyderabad Excellence in Research Award. I have industry research experience at Dolby, Microsoft Research, and Adobe Research, with expertise spanning multimodal and generative AI, efficient inference, out-of-distribution robustness, content protection and attribution, and model evaluation. My work has been published at CVPR, WACV, ECCV, NeurIPS, and in IEEE Transactions on Image Processing, and I have experience developing and benchmarking large language models, multimodal systems, diffusion models, and real-world computer vision prototypes.

Research and Industry Experience

Senior Multimodal AI Researcher — **Sep. 2025 – Present**
Dolby Advanced Technology Group India — *Bengaluru, India*

- Developing media-level watermarking methods for piracy deterrence in live-streaming content across offline and online settings; contributing to an invention currently being prepared for patent filing.
- Designed and executed a content-attribution study for text-to-image diffusion models, evaluating SAM3, DINO, and CLIP-based approaches on Stable Diffusion 1.5 and Stable Diffusion XL using precision and recall.
- Building a robust evaluation framework for generated multimodal content across **5 open-source models** and **20+ evaluation dimensions**, using benchmarks with human-alignment annotations to ground and validate model-generated scores.

Research Intern — **Jun. 2025 – Aug. 2025**
Adobe Research India — *Bengaluru, India*

- Built a proof-of-concept **layer-wise text-to-poster generation and editing pipeline** that used a multimodal LLM to reason over input text and existing poster elements before applying targeted modifications.
- Delivered an interactive demo and conducted side-by-side qualitative comparisons with GPT-4o. On the evaluated poster examples, the prototype showed stronger instruction adherence and fewer unintended edits.

Research Intern — **Jul. 2024 – Dec. 2024**
Microsoft Research India — *Bengaluru, India*

- Built and benchmarked a **tree-based speculative-decoding** method for **Vicuna-7B** and **Llama-3-Instruct-8B** on an NVIDIA A100 80GB GPU, achieving speedups of 2.91× and 2.62×, with 4.08 and 4.12 accepted tokens per target-model forward pass, respectively.
- Matched with benchmark EAGLE's Vicuna-7B speedup (2.91× vs. 2.90×) and improved accepted length over EAGLE on Vicuna-7B (4.08 vs. 3.94) and Llama-3-Instruct-8B (4.12 vs. 3.65), while characterizing throughput–acceptance trade-offs against EAGLE and EAGLE-2.
- Designed an **OOD evaluation** study for **Phi-3.5-Vision and LLaVA-OneVision** across 8 domains of OfficeHome and TerraIncognita datasets. Quantified in-distribution/OOD degradation and evaluated prompt optimization and adversarial LoRA strategies, identifying limitations of the tested interventions under domain shift.

Research Intern — **Mar. 2023 – Nov. 2023**
Autonomous and Cyber-Physical Systems Lab, University of Agder — *Grimstad, Norway*

- Developed adversarial-learning methods for domain generalization and multi-source domain adaptation, including theoretical analysis using unseen-target error bounds; the work resulted in first-author publications at **NeurIPS 2023** and **IEEE TIP 2025**.

Visiting Researcher — **Jan. 2020 – Jul. 2021**
Autonomous and Cyber-Physical Systems Lab, University of Agder — *Grimstad, Norway*

- Contributed to **six published interdisciplinary deep-learning projects** spanning biomedical imaging, sensor-based systems, and autonomous vehicles.
- Optimized and profiled lightweight models for **edge deployment** on NVIDIA Jetson AGX Xavier and Raspberry Pi 4B using TensorFlow Lite and ONNX, focusing on inference latency and memory footprint.
- Developed a real-time surround-view ADAS prototype that fused camera imagery and 3D point clouds for blind-spot reduction and object-distance estimation.

Education

Indian Institute of Technology Hyderabad — **Aug. 2021 – Dec. 2025**
Ph.D. in Artificial Intelligence, GPA: 9.19/10 — Hyderabad, India
Prime Minister's Research Fellow (PMRF); Excellence in Research Award, 2024

BML Munjal University — **May 2020**
B.Tech. in Computer Science and Engineering, GPA: 9.44/10 — Gurgaon, India

Selected Publications

- **A. Dayal**, P. Divya, N. Tiwari, L. R. Cenkeramaddi, C. K. Mohan, and A. Kumar, "Bridging the domain gap in small multimodal models: A dual-level alignment perspective," ***IEEE/CVF WACV***, 2026.
- S. VCR, R. Lalla, **A. Dayal**, T. Kulkarni, A. Lalla, V. N. Balasubramanian, and M. H. Khan, "Foundation model priors enhance object focus in feature space for source-free object detection," ***IEEE/CVF CVPR***, 2026.
- **A. Dayal**, Shrusti S., L. R. Cenkeramaddi, C. K. Mohan, and A. Kumar, "Leveraging mixture alignment for multi-source domain adaptation," ***IEEE Transactions on Image Processing***, 2025.
- **A. Dayal**, R. Lalla, L. R. Cenkeramaddi, C. K. Mohan, A. Kumar, and V. N. Balasubramanian, "Improving unsupervised domain adaptation: A pseudo-candidate set approach," ***ECCV***, 2024.
- **A. Dayal**, V. K. B., L. R. Cenkeramaddi, C. K. Mohan, A. Kumar, and V. N. Balasubramanian, "MADG: Margin-based adversarial learning for domain generalization," ***NeurIPS***, 2023.
- **A. Dayal**, M. Aishwarya, S. Abhilash, C. K. Mohan, A. Kumar, and L. R. Cenkeramaddi, "Adversarial unsupervised domain adaptation for hand gesture recognition using thermal images," ***IEEE Sensors Journal***, 2023.
- **A. Dayal**, L. R. Cenkeramaddi, and A. Jha, "Reward criteria impact on the performance of reinforcement learning agent for autonomous navigation," ***Applied Soft Computing***, 2022.
- **A. Dayal**, S. R. Yeduri, B. H. Koduru, R. K. Jaiswal, J. Soumya, M. B. Srinivas, O. J. Pandey, and L. R. Cenkeramaddi, "Lightweight deep convolution neural network for background sound classification in speech signals," ***Journal of the Acoustical Society of America***, 2022.

Selected Applied AI Projects

Computer-Vision System for Aircraft Movement in Hangars *Industry collaboration with Innominds*

- Co-developed a proof of concept that fused DeepLabV3 semantic segmentation from RGB cameras with LiDAR point clouds and implemented collision-avoidance logic to assist aircraft movement inside hangars.

Autonomous Drone-Based Medical Delivery System *IIT Hyderabad BUILD initiative*

- Co-developed a DJI Tello navigation pipeline using its onboard camera feed to detect spatial QR-code destinations and execute autonomous landing for a medical-delivery proof of concept.

Real-Time Surround-View System for ADAS *Bachelor's thesis / University of Agder*

- Built and deployed a 360-degree surround-view and object-distance estimation system on a TurtleBot3 by fusing 2D camera imagery with 3D point clouds for blind-spot reduction and collision avoidance.

Multi-Object Detection and Color Identification *End-to-end IoT prototype*

- Developed an assistive-vision device using YOLOv2 and K-means clustering, achieving **76.8 mAP** for object detection and **85.2% accuracy** for color recognition.

Technical Skills

Programming and ML: Python, PyTorch, TensorFlow, scikit-learn, NumPy, OpenCV
Models and Methods: Hugging Face Transformers, multimodal models, diffusion models, parameter-efficient fine-tuning, prompt optimization
Systems and Evaluation: Distributed training/inference, profiling, APIs and model serving, Linux, Git, experiment tracking, OOD and human-aligned evaluation

Selected Honors and Recognition

- Awarded the **Excellence in Research Award 2024** by IIT Hyderabad; selected as one of 28 students across 15 departments.
- Awarded the **Prime Minister's Research Fellowship (PMRF)** in 2022.
- Received the **Microsoft Research Travel Grant** to attend NeurIPS 2023 in the United States and to attend ECCV 2024 in Italy.
- Invited to present research at the **3rd INMOST Workshop** on Indo-Norwegian Collaboration in Intelligent Offshore Mechatronics Systems, University of Agder, Grimstad, Norway.
- Invited to deliver multiple **talks on the Foundations of Machine Learning** at several institutes, Vardhaman College of Engineering, B.V. Raju Institute of Technology and Bharat Institute of Engineering and Technology.

Academic Service

Served as a reviewer for **CVPR, ICCV, ECCV, NeurIPS, ICML, and ICLR**, as well as several reputed journals.